

Seat Majorities without Vote Majorities: Coalition Inversions in Brazil’s Chamber of Deputies

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Abstract

Proportional representation is usually evaluated party by party, by comparing each party’s share of seats with its share of votes. Yet fragmented legislatures are governed through combinations of parties, so small party-level deviations can become consequential when parties are combined. Brazil provides a useful setting because the Chamber of Deputies is assembled from twenty-seven separate district elections. Several institutional steps separate national votes from final party seat totals, and no national compensatory allocation reconciles them. I ask whether politically meaningful sets of parties can hold a legislative majority despite receiving less than half of all valid federal-deputy votes nationwide, a condition known as coalition inversion. Using the 2014, 2018, and 2022 elections, I compare distinct election-year cabinet party sets with potential coalitions restricted by ideological proximity. Cabinet inversions appear following the 2014 and 2022 elections, but not the 2018 election; inversions among minimal connected winning coalitions appear in all three. In 2022, a seven-party bloc of ideologically adjacent parties combined 45.26 percent of the vote with 258 of 513 seats. Party-level deviations from proportionality can therefore have a distinct coalition-level consequence: they can determine which combinations of parties command a legislative majority.

1 Introduction

Under proportional representation, vote–seat translation is usually assessed party by party, by comparing each party’s vote and seat shares (Kaminski 2026). In fragmented legislatures, however, governing and legislative majorities are assembled from several parties. Combining parties can make individually modest deviations consequential. Positive and negative deviations may cancel or reinforce one another, allowing a party set to cross the legislative majority threshold before its combined vote share reaches one half. This configuration is a coalition inversion: a party set holds a seat majority while receiving less than half of all valid national legislative votes.

Kurrild-Klitgaard (2013) documents empirical cases of coalition inversions in Danish government formation. Miller (2014) establishes the more general result that inversions remain possible even when seats are allocated in a single national constituency, without an explicit seat threshold, and by a highly proportional formula. Because seats must be assigned in whole numbers, party seat shares will generally depart slightly from exact proportionality. Once positive and negative party-level deviations exist, some combinations of parties can reverse the vote and seat majorities. These results settle the existence question. They do not show how coalition inversions appear when national party seat totals are themselves assembled from many district elections.

Brazil provides such a setting. The Chamber of Deputies is not formed by applying one proportional allocation to national party votes. Its seats are allocated separately across the twenty-six states and the Federal District. District magnitudes and electoral weights vary, electoral rules and the units over which votes were pooled changed across the three elections, and party support is distributed unevenly across the country. Several institutional and geographical mediations therefore stand between a party’s national vote share and its final national seat share (Bochsler 2026).

This institutional structure changes the sources of deviation without changing the national proportionality benchmark. I compare seat shares with shares of all valid federal-deputy votes, including votes for parties that obtain no seats. Restricting the denominator to votes for seat-winning parties would instead evaluate representation conditional on which parties entered the Chamber. The question here concerns the national outcome of the entire vote-to-seat process, including electoral exclusion. Such exclusion can affect the distribution of seats among represented parties, as work on spoiler effects under proportional representation demonstrates (Kaminski 2018). The benchmark therefore measures departures from national proportionality without presuming that those departures are unintended or unjustified.

Party-level deviations emerge from twenty-seven district-level translations operating under heterogeneous conditions, rather than only from rounding a single national allocation. The organizing question is therefore: when do these deviations combine so that a politically meaningful set of parties holds a legislative majority without a national vote majority? The contribution lies in showing how the resulting party-level seat differentials aggregate across coalitions and change majority status.

The Chamber contains many mathematically possible party subsets, most with little political meaning. The first empirical domain therefore uses realized cabinet coalitions. Changes in ministerial participation generate successive party sets during a legislative term; for each, I compare the Chamber seats of its members with their votes in the corresponding election.

The second domain considers potential coalitions restricted by ideological proximity. The baseline is the set of minimal connected winning coalitions: contiguous blocs in the election-year left–right order that hold a seat majority and lose it if narrowed (Axelrod 1970). A one-gap check permits one represented party inside the span to be omitted. For both domains, I analyze the coalition seat differential and its decomposition into within-district and between-district components.

Inversions appear in both coalition domains, but their distribution differs across elections. Two of the 35 distinct cabinet party sets invert: one observed for five days under Dilma Rousseff following the 2014 election and one observed during the first 255 days of Lula III following the 2022 election. No cabinet party set linked to the 2018 election inverts in the primary chronology. Minimal connected winning inversions, by contrast, appear in all three elections. The 2022 cases include a compact right-side bloc with 45.26 percent of the vote and 258 seats. Within-district and between-district components reinforce one another in some inversions and offset one another in others.

The remainder of the article is structured as follows. Section 2 describes Brazil’s electoral system; Section 3 defines the measures and coalition domains; Section 3.2 presents the data; the remaining sections report and discuss the results.

2 Brazil’s multidistrict open-list proportional system

Coalition majorities are formed from national party seat totals, but those totals are not produced by a national allocation. In Brazil, elections to the Chamber of Deputies take place separately in

the twenty-six states and the Federal District, each of which forms an electoral district. The 513 seats are allocated within these twenty-seven districts and then summed nationally. The Federal District is simply an eight-seat district, and no compensatory tier subsequently adjusts the resulting party totals, unlike compensatory mixed systems, where a second allocation stage can correct constituency-level disproportionality (Flis, Grofman, and Kaminski 2025; Salamon, Kaminski, and Flis 2026).

District magnitude varies considerably across states, from eight to seventy seats. Eleven districts elect only eight deputies, the median magnitude is ten, the mean nineteen, and São Paulo elects seventy. These differences imply substantial variation in the effective conditions of representation across districts: winning a seat is generally more demanding where fewer seats are available. Because each district is allocated independently, the resulting advantages and shortfalls carry into national party seat totals.

Within each district, voters encounter an open-list election. They may enter the identifying number of an individual candidate or cast a vote for a party label. A candidate vote performs two functions at once. It contributes to the aggregate vote of the party, electoral coalition, or federation through which the candidate competes, and it raises that candidate’s position within the relevant open list. A party-label vote contributes to the first calculation but not to the ranking among candidates. That is, votes first determine how many seats an electoral list receives. Nominal votes then determine which candidates within that list take the seats, subject to the individual eligibility requirements in force for the election. A candidate can therefore benefit from votes cast for copartisans or electoral partners, but cannot ordinarily occupy a seat unless the list to which the candidate belongs wins one (Nicolau 2006).

Valid candidate and party-label votes are first aggregated for the electoral unit in force in district d , whether a party, proportional coalition, or federation. The electoral quotient and party quotient are $QE_d = \text{round}_{\text{stat}}\left(\frac{V_d}{M_d}\right)$ and $QP_{\ell d} = \left\lfloor \frac{v_{\ell d}}{QE_d} \right\rfloor$. The party quotient determines the initial list seats, subject to the candidate-eligibility rules applicable in that election (Brasil 1965, arts. 106–108).

sequentially by highest averages, $v_{\ell d}/(s_{\ell d} + 1)$, producing the Jefferson–D’Hondt sequence $v_{\ell d}/1, v_{\ell d}/2, \dots$ until all seats are filled. Eligibility for these remainder seats follows the year-specific rules described below (Brasil 1965; Nicolau and Schmitt 1995, art. 109).

Eligibility rules changed across elections. In 2014, there was no fixed candidate threshold, but only lists reaching the electoral quotient could contest remainders (Tribunal Superior Eleitoral 2013, art. 186, §2). In 2018, candidates ordinarily needed 10 percent of the quotient; all lists could contest remainders while an eligible candidate remained, after which any still-unfilled seats were assigned by highest averages without that nominal-vote minimum (Tribunal Superior Eleitoral 2017, arts. 7–10). In 2022, the 10 percent candidate threshold remained for initial seats. For remainder seats, lists first needed 80 percent of the quotient and candidates 20 percent; any seats still unfilled were then assigned by highest averages among lists meeting the 80 percent requirement, without the 20 percent candidate threshold (Tribunal Superior Eleitoral 2021b, arts. 8–11).

Moreover, proportional electoral coalitions, or *coligações proporcionais*, were part of the 2014 and 2018 elections. They were formed within the electoral circumscription, which for the Chamber meant a state or the Federal District. A party could consequently enter different proportional arrangements in different states, subject to the legal restrictions in force for the election. These were not national alliances whose composition automatically carried across all twenty-seven contests (Nicolau 2006; Tribunal Superior Eleitoral 2013, 2017). Member parties of proportional electoral coalitions legally operated as a joint open list when votes were converted into seats. Candidate votes and party-label votes credited to any member entered the coalition’s district total; the quotient and highest-averages calculations then assigned seats to the coalition as a whole. Those seats were

occupied by the highest nominal vote-getters across the combined list, regardless of the member party to which they belonged. A candidate from one party could therefore be elected partly because votes cast for candidates or labels of another coalition member raised the joint total (Nicolau 2006; [Tribunal Superior Eleitoral 2013, 2017](#)).

This resembles list apparentement because parties pool votes before seats are settled, but the Brazilian arrangement differs from the two-stage system studied by Pukelsheim and Leutgäb (2013). There, alliance seats are subsequently apportioned among component lists. In Brazil, a single joint open list selected winners by candidate votes, so gains to particular members depended on the distribution of nominal votes rather than on a fixed secondary apportionment rule.

Constitutional reform abolished proportional electoral coalitions, and party federations were subsequently introduced as a new form of multiparty organization for proportional elections. Three federations competed in 2022: PT–PCdoB–PV, PSDB–Cidadania, and PSOL–REDE ([Tribunal Superior Eleitoral 2026](#)). Like the former *coligações*, federations pooled their members’ votes in a common open list, so constituent-party seat totals resulted from competition within the pooled list rather than from independent proportional allocations.

Unlike the former electoral coalitions, federations operate nationally and in principle persist for at least four years. They have some common congressional procedures, including leadership and committee-allocation arrangements, but do not merge their constituent parties, which retain separate identities and organizational autonomy ([Tribunal Superior Eleitoral 2021a](#)).

National proportionality provides a common baseline against which to assess how Brazil’s layered vote-to-seat process generates representational deviations and when those deviations become majority-changing as parties are recombined into coalitions. If v_i is the national valid vote total credited to party i and V is the corresponding total for all parties, including those that win no seats, the party’s reference quota is $513v_i/V$ seats. Brazil’s actual system never performs this national allocation.

Several mediations therefore stand between a party’s national vote share and its national seat share. Within each district, finite magnitude, the quotient and highest-averages calculations, legal eligibility thresholds, and the number and relative strength of competing electoral units shape the conversion of votes into seats. Electoral pooling adds a further mediation by changing both the vote total on which seats are allocated and, through the common candidate list, the eventual distribution of those seats among member parties. These arrangements varied across districts in 2014 and 2018 and became nationally fixed through federations in 2022.

The national seat distribution then aggregates twenty-seven independently determined district outcomes. Chamber seats are apportioned by population, but the constitutional floor of eight and ceiling of seventy prevent exact population-to-seat equality across states (Samuels and Snyder 2001; [Brasil 1993](#)). Turnout and valid-vote rates also differ across elections and districts, so a seat can correspond to different numbers of valid votes in different states. Because parties distribute their support unevenly across the country, national vote shares can therefore be assembled from geographically different configurations with different representational consequences. Electoral geography interacts with district seat weights and local competitive conditions before the resulting party seats are summed nationally. The resulting disproportionality need not have a single source: allocation rules, district magnitude, malapportionment and electoral weight, electoral geography, and party competition constitute distinct channels through which party vote and seat shares can diverge (Bochsler 2026).

This layered translation occurs in an unusually fragmented party environment. Measured by the Laakso–Taagepera index, the effective numbers of electoral parties were 14.11 in 2014, 18.02 in 2018, and 12.34 in 2022; the corresponding effective numbers of parliamentary parties were

13.33, 16.41, and 9.91 (Laakso and Taagepera 1979). The exact number of allocation units was not identical to the number of party labels, especially when district coalitions or national federations pooled votes. Even so, each election produced a Chamber in which legislative majorities had to be assembled across numerous partisan units. Fragmentation makes the political consequences of modest party-level gains and losses depend on how those parties are combined.

The national party seat totals produced by this process enter a presidential system in which governing capacity ordinarily depends on multiparty arrangements. Brazilian coalitional presidentialism links the presidency to combinations of parties across cabinet and Congress: ministries are distributed among several parties, while executive–legislative coordination is organized through agreements among them (Abranches 1988; Figueiredo and Limongi 2000). The Chamber assembled from twenty-seven district elections therefore supplies the parliamentary resources from which governing coalitions are constructed. Party-level representation is politically consequential not only in isolation, but also through the combinations of parties that can command a legislative majority.

The realized party seat totals incorporate the effects of rounding, highest averages, district magnitude, eligibility requirements, electoral pooling, apportionment, turnout, valid-vote variation, electoral geography, and district-level competition. Comparing those totals with national proportional quotas identifies the positive and negative representational deviations carried by each party.

3 Coalition inversions, domains, and measurement

Party seat totals need not equal their national proportional quotas. When parties are combined, their positive and negative differentials may cancel or reinforce. A coalition inversion occurs when the accumulated differential carries a party set above the Chamber’s 257-seat threshold before its combined vote reaches one half of the national total. The possibility of coalition inversions under proportional representation is already established by Kurrild-Klitgaard (Kurrild-Klitgaard 2013) and Miller (Miller 2014). The present analysis extends this agenda to a distinct institutional setting, where national party seat totals emerge from twenty-seven independent district allocations and several additional electoral mediations, and to coalition domains defined by realized cabinet participation and ideological proximity.

Because a fragmented legislature permits many mathematically possible party sets, I restrict the coalition space in two ways. Observed cabinets identify realized governing combinations; ideological connectedness defines a structured domain of potential coalitions. The two domains show, respectively, how party-level differentials combine in government and where potential inversions appear in the ideological ordering.

Cabinet coalitions are substantively meaningful because Brazilian presidents ordinarily govern through multiparty arrangements. Cabinet participation records a party combination that was not merely arithmetically possible but institutionally realized in executive government. The relevant representational question is, thus, whether the members of the party set holding ministries collectively possessed a Chamber majority despite receiving less than half of the valid federal-deputy vote in the election that produced that legislature.

Cabinet membership changes during a legislative term while the election-year vote and seat vector remains fixed. These transitions recombine the same party-level differentials, allowing inversion status to change without a new electoral allocation. The cabinet sequence therefore shows whether, and for how long, realized governing sets cross the seat-majority threshold without a vote majority.

Observed cabinets cover only a small share of possible parliamentary majorities, so I also

examine potential coalitions structured by ideological proximity. Following Axelrod (1970), I use minimal connected winning coalitions: contiguous intervals in the election-year ideological order that hold at least 257 seats and lose that majority when narrowed. This domain identifies where ideologically contiguous party sets first acquire a seat majority. The inversion question can then be asked directly at that frontier: which of these ideologically coherent seat majorities still fall short of a national vote majority?

3.1 Measurement

Let P denote the set of all parties receiving valid votes in a federal-deputy election, including those that win no seats. For each party $i \in P$, let v_i be its national total of valid candidate and party-label votes and s_i its Chamber seats. With $V = \sum_{i \in P} v_i$ denoting the total valid federal-deputy vote and $S = 513$, exact national proportionality assigns party i the quota

$$q_i = S \frac{v_i}{V}.$$

The party seat differential is $d_i = s_i - q_i$; positive values denote more seats than the national quota and negative values fewer. Because seats and quotas each sum to S , party differentials sum to zero. For coalition $C \subseteq P$,

$$v_C = \sum_{i \in C} v_i, \quad s_C = \sum_{i \in C} s_i, \quad q_C = S \frac{v_C}{V}, \quad d_C = s_C - q_C = \sum_{i \in C} d_i.$$

The coalition's departure from exact national proportionality is produced by adding the positive and negative differentials of its member parties. A party with a negative contribution can belong to an inverted coalition, provided that the positive contributions elsewhere in the set more than offset it.

A coalition inversion occurs when the party set receives less than half of the national valid federal-deputy vote but controls at least 257 Chamber seats: $\frac{v_C}{V} < \frac{1}{2}$ and $s_C \geq 257$. For a vote-minority coalition, the seat-majority condition is $d_C \geq 257 - q_C$: its accumulated differential must bridge the distance from its national proportional quota to 257 seats. Because C and its complement partition both votes and seats, an inversion simultaneously gives the complement a vote majority and at most 256 seats. Throughout, V includes all valid federal-deputy votes, including votes for parties winning no seats.

Representation ratios express the same departures in relative rather than absolute terms (Taagepera and Laakso 1980). At the party and coalition levels,

$$R_i = \frac{s_i/S}{v_i/V} = \frac{s_i}{q_i}, \quad R_C = \frac{s_C/S}{v_C/V} = \frac{s_C}{q_C} = 1 + \frac{d_C}{q_C} = \sum_{i \in C} \frac{q_i}{q_C} R_i.$$

A value of one denotes exact proportionality, a value above one denotes overrepresentation, and a value below one denotes underrepresentation. Parties receiving votes but no seats have $R_i = 0$. The ratios permit comparisons across parties and coalitions of different sizes. The differentials remain the more direct quantities for the majority-threshold question because they measure the relevant departure in seats.

The national coalition differential can be decomposed by adding and subtracting the seats that exact proportionality would assign to the coalition separately within each district:

$$d_C = \left[\sum_d \left(s_{Cd} - S_d \frac{v_{Cd}}{V_d} \right) \right] + \left[\left(\sum_d S_d \frac{v_{Cd}}{V_d} \right) - \left(S \frac{v_C}{V} \right) \right].$$

The first term compares the coalition’s actual seats in each district with the number implied by its vote share in that district. I denote it the within-district component, A_C . The other component is the between-district component, B_C . Its first term sums the coalition’s proportional quota in each district. The second is the quota implied by the coalition’s share of the national valid vote. These quantities need not coincide because districts differ in the number of seats they contribute relative to the number of valid votes cast in them. Thus, B_C depends on the geographic distribution of coalition support across districts with different electoral weights. Support concentrated in districts whose share of Chamber seats exceeds their share of the national valid vote raises the sum of district quotas relative to the national quota, while support concentrated in districts with the opposite weighting lowers it.

Thus, $d_C = A_C + B_C$. For comparisons across coalitions of different electoral sizes, the two components can also be expressed relative to the coalition’s proportional quota:

$$R_C - 1 = \frac{A_C}{q_C} + \frac{B_C}{q_C}.$$

This expresses the within- and between-district contributions to the coalition’s relative departure from proportionality.

In Brazil, A_C can reflect the mechanisms that operate within each district when votes are converted into seats: finite district magnitude, the electoral-quotient and highest-averages calculations, the year-specific eligibility rules for parties and candidates, and the pooling of votes through proportional electoral coalitions in 2014 and 2018 and party federations in 2022. These mechanisms determine how the coalition’s actual district seats depart from its district-level proportional quotas. By contrast, B_C arises from differences in electoral weight across districts and the geographic distribution of coalition support: district apportionment, turnout, and valid-vote variation determine how many seats correspond to a given number of votes in each district, while coalition geography determines whether its support is concentrated in relatively more or less heavily weighted districts. Neither component isolates the independent effect of any one mechanism; the decomposition locates the coalition differential within or between districts rather than providing a causal decomposition of its institutional sources (Bochsler 2026).

The primary ideological universe is the set of parliamentary parties,

$$P^+ = \{i \in P : s_i > 0\}.$$

National proportionality is evaluated over P , while potential parliamentary coalitions are constructed over P^+ . Thus, votes for parties without seats remain part of V and of party-level proportionality accounting, but those parties cannot constitute or interrupt a primary ideological coalition. This restriction does not alter observed cabinet membership or its election-year translation.

Axelrod’s connected-coalition criterion provides the baseline restriction (Axelrod 1970). Let $p_1, \dots, p_n$ denote the existing election-year ideological order restricted to P^+ , preserving the relative ordering of represented parties without re-estimating their positions. A coalition $C \subseteq P^+$ is connected when its members form a contiguous interval in this filtered order. A connected coalition is minimal winning when $s_C \geq 257$ and no proper connected subset retains a Chamber majority. For an interval, removing either endpoint must therefore leave fewer than 257 seats.

Exact connectedness ($k = 0$) provides the baseline ideological domain. As a nested sensitivity analysis, I also allow one represented party inside a coalition’s ideological span to be omitted ($k = 1$). The formal definition and summary results for this one-gap domain appear in Appendix B.1; full coalition-level results are in the machine-readable replication outputs.

As a sensitivity analysis, I repeat both searches in the full election-year order over P , including parties receiving votes but no seats. In that universe, such parties can be coalition members and can interrupt connectedness. Admissibility and minimality are recomputed within that order. Only the ideological universe changes: votes, seats, quotas, differentials, representation ratios, and the decomposition retain the same definitions and national vote denominator. Appendix B.2 compares the results.

3.2 Data and operationalization

Vote and seat data for the 2014, 2018, and 2022 federal-deputy elections come from the Tribunal Superior Eleitoral, accessed through the `electionsBR` package (Meireles, Silva, and Costa 2016).¹ For each party, the national vote total combines valid candidate votes and party-label votes across the twenty-seven districts, while the seat total counts deputies elected under its election-year label.² This attribution also applies to parties that competed through proportional electoral coalitions in 2014 and 2018 or federations in 2022. The resulting party totals connect the district electoral outcomes to the party identities observed in cabinet participation and ideological classifications.

Cabinet composition is reconstructed from publicly available online records for the Dilma Rousseff, Michel Temer, Jair Bolsonaro, and Lula III administrations, recording each officeholder’s service dates and party affiliation at the time. I map these dated affiliations to the party labels of the election that produced the corresponding legislative term and treat each distinct party set within that term as one cabinet configuration. When the same configuration recurs, its dates are retained for duration summaries but it is not counted as a new configuration.

Each cabinet configuration is evaluated against the vote and seat distribution of the election that produced the corresponding Chamber: 2014 for the 2015–2018 term, 2018 for 2019–2022, and 2022 for Lula III through March 19, 2026. Parties that renamed or merged during a term are matched to the election-year parties from which their vote and seat totals derive. This yields 12 distinct cabinet configurations for the 2014 term, 18 for 2018, and 5 for 2022. Appendix A.1 reports their membership.

Cabinet affiliations are identified for 3,996 of the 4,096 days covered by the chronology. The remaining 100 days involve unresolved affiliations for a small number of officeholders. Alternative codings for the recorded cases do not change which cabinet configurations invert or the 260 days for which an inversion is observed.

The ideological orders for 2014 and 2018 use the party classifications of Bolognesi, Ribeiro, and Codato (2023). For 2022, I use the later 2018–2022 classification, drawing on the 2025 data release and associated article (Bolognesi et al. 2026). Each classification supplies the election-year party order used to construct the coalition domains defined above.

4 Results

Before turning to coalitions, I first examine the party-level seat differentials d_i and representation ratios R_i produced in each election, which describe the absolute and relative departures from national proportionality that are subsequently combined at the coalition level.

In the 2014 election PMDB received 8.13 seats above its national proportional quota, the largest positive differential that year. PSD and PTB received another 4.55 and 4.37 seats above their

1. All 2022 results refer to the official seat vector in the frozen dataset, prior to subsequent judicial retotalizations.

2. For 2014, the vote measure is `QT_VOTOS_NOMINAIS` + `QT_VOTOS_LEGENDA`. For 2018 and 2022, it is `QT_VOTOS_NOMINAIS_VALIDOS` + `QT_TOTAL_VOTOS_LEG_VALIDOS`.

quotas, respectively. These gains coexisted with deficits of 4.43 seats for PSDB and 4.20 for PSOL.

In 2018, PP, PR, and MDB had positive differentials of 8.76, 5.72, and 5.60 seats, while PSL and NOVO fell 7.70 and 6.35 seats below quota. PSL had the largest national vote total but the largest negative differential, showing that party size affects the magnitude of representation without fixing its sign. Whether a party is over- or underrepresented nationally depends on both its vote-to-seat conversion within districts and the distribution of its votes across districts with different electoral weights.

In 2022, positive differentials were concentrated among large parties: PL, UNIÃO, PT, and PP gained 13.78, 11.10, 6.94, and 6.24 seats above quota. Because $d_i = q_i(R_i - 1)$, electoral size scales the seat consequence of a given relative departure from proportionality. Large parties lying on the positive side therefore placed substantial surpluses inside many available coalition combinations.

Figure 1 shows greater dispersion in representation ratios among small parties, while larger parties generally lie closer to $R_i = 1$. The direction of the deviation does not follow a clear size pattern in 2014 or 2018. PT and PSDB were underrepresented in 2014 while PMDB, of comparable size, was overrepresented; in 2018, the largest party, PSL, was underrepresented while several smaller parties were overrepresented. The 2022 profile is different: every party receiving more than 5 percent of the national vote was overrepresented, while every party between 3 and 5 percent was underrepresented. A clear association between larger size and overrepresentation therefore appears only in 2022.

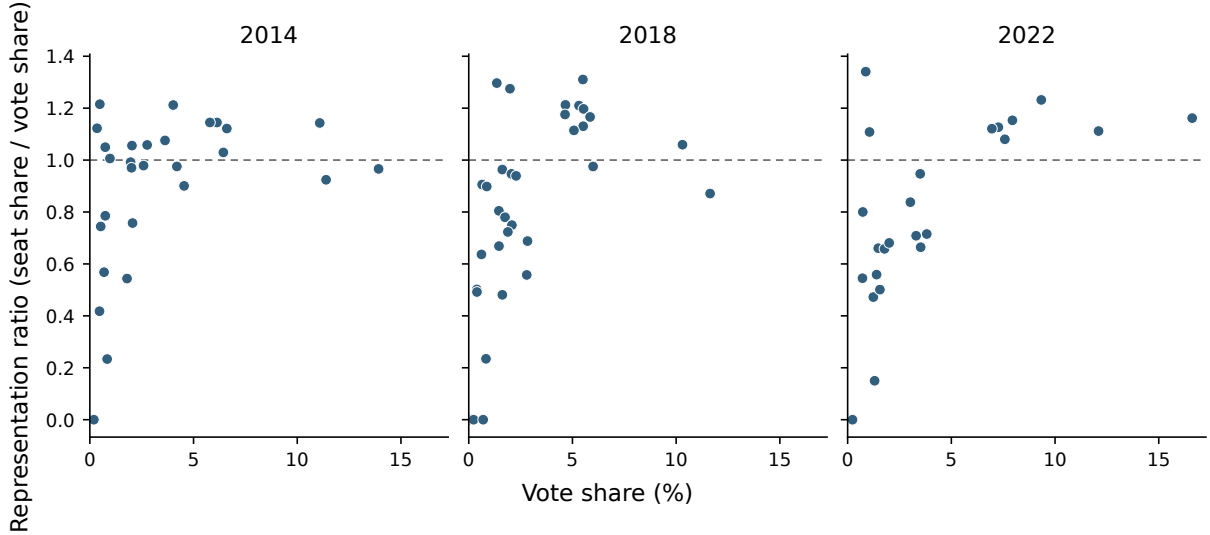


Figure 1: Party-level representation profiles, 2014, 2018, and 2022

Notes: Each point is a party. $R_i = 1$ denotes exact proportionality; parties receiving votes but no seats appear at zero.

Representation ratios alone also do not identify the largest contributors to a coalition surplus because R_C is quota-weighted and $d_C = \sum_{i \in C} d_i$. In 2022, PV had the higher ratio ($R_i = 1.341$) but only a 1.52-seat surplus, whereas PL's lower ratio of 1.162 produced 13.78 surplus seats. Majority status therefore depends on the size-weighted combination of member gains and losses and on whether their total bridges the distance to 257 seats.

4.1 Cabinet coalitional inversions

Two of the 35 distinct cabinet party sets combine a Chamber majority with less than half of the national federal-deputy vote. They occur under Dilma Rousseff and Lula III, following the 2014 and 2022 elections, respectively. No set observed under Michel Temer or Jair Bolsonaro inverts in the primary chronology. The two inversions differ in their distance from a vote majority, the size of their seat advantage, and their observed duration. Table 1 reports these sets and the within-district and between-district components of their differentials.

Table 1: Inverted cabinet party sets and their district components

Election	Set	Days	Vote %	Seats	d_C	A_C	B_C	Parties
2014	14-05	5	46.54	259	20.271	15.742	4.529	PCdoB, PDT, PMDB, PR, PSD, PT, PTB
2022	22-01	255	48.89	263	12.204	13.196	-0.992	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO

Days sum actual appearances of each set; all inversion days are evidence-established.

Notes: Votes, seats, and party labels refer to the corresponding election. A_C and B_C are the within-district and between-district components.

The first inversion emerges during the contraction of Dilma Rousseff’s cabinet in April 2016. PRB ceases to be represented on March 18, and PP’s departure on April 14 leaves a party set with 259 seats and 46.54 percent of the 2014 vote. Exact national proportionality would assign those parties 238.73 seats, leaving them 18.27 seats short of a Chamber majority. Their 20.27-seat differential, corresponding to $R_C = 1.085$, is therefore sufficient to preserve the majority. Set 14-05 is observed for five days, from April 14 through April 18. PSD’s departure on April 19 reduces the party set to 223 seats and ends the inversion.

The Temer sequence does not produce an inversion. The cabinet party set observed from April 2 through April 5, 2018 comes close, with 268 seats and 50.03 percent of the 2014 vote, but it still has a vote majority. From April 6, the remaining parties have 247 seats and 45.83 percent of the vote. The contraction thus moves from a majority in both measures to a minority in both, rather than leaving a vote-minority seat majority. A positive seat differential is not, by itself, evidence of an inversion.

The cabinet sequence linked to 2018 also contains no inversion. Set 18-11 holds 222 seats with 39.79 percent of the vote. Its quota is 204.14 seats and its 17.86-seat surplus gives $R_C = 1.087$, but the set remains 52.86 seats short of the majority threshold. It is substantially overrepresented without being close to inversion.

The second cabinet inversion appears at the beginning of Lula III. Its members hold 263 seats with 48.89 percent of the 2022 vote. Their proportional quota is 250.80 seats, so the 12.20-seat differential ($R_C = 1.049$) is sufficient to bridge the 6.20-seat gap to a majority. The coalition therefore holds a larger seat majority than the Dilma inversion despite a smaller differential, because it begins much closer to the 257-seat threshold. This configuration persists for 255 days, until PP and Republicanos enter the cabinet on September 13, 2023. Their entry raises the party set to 350 seats and 63.79 percent of the vote, ending the inversion by adding a vote majority rather than by removing the seat majority, as occurred in the Dilma case.

The positive within-district component of the two inversions extends to all 35 cabinet configurations: $A_C > 0$ in every case, while $B_C > 0$ in only thirteen. Cabinet composition helps explain this regularity. In every configuration, cabinet members have a higher mean national vote

share than parties outside the cabinet in the same election-year party system, with gaps ranging from 1.46 to 4.95 percentage points. Larger parties are also usually on the positive side of the within-district component: among parties receiving at least 5 percent of the national vote, $A_i > 0$ for six of seven in 2014, eight of nine in 2018, and all seven in 2022. In 34 of the 35 cabinet configurations, the positive A_i contributions of members above this benchmark exceed all negative member contributions combined; only one configuration also requires positive contributions from smaller members to keep $A_C > 0$. Cabinet composition therefore repeatedly brings together larger parties whose within-district contributions are predominantly positive. No comparable sign pattern appears for B_C . of coalition support across districts.

The two inverted cabinet party sets differ in how the positive A_C interacts with between-district weighting. For 14-05, $A_C = 15.74$ raises the coalition’s quota of 238.73 to $q_C + A_C = 254.47$ seats, still short of a Chamber majority. Adding $B_C = 4.53$ brings the observed total to 259 seats. For 22-01, by contrast, $A_C = 13.20$ already exceeds the 6.20 seats separating the quota from a majority: $q_C + A_C = 263.99$. The negative between-district component, $B_C = -0.99$, reduces the total to 263 seats without removing the majority. Thus, within-district gains characterize both inversions, while between-district weighting reinforces the Dilma coalition and offsets the Lula coalition. Only in the former is the positive between-district component also needed to bridge the proportional distance to 257 seats.

Overrepresentation is much more common than inversion: 33 of the 35 cabinet sets have $R_C > 1$, while only two invert. Figure 2 shows each distinct set once, ordered by first appearance; it does not encode duration or recurrence. Most overrepresented sets remain either below both majority thresholds or above both.

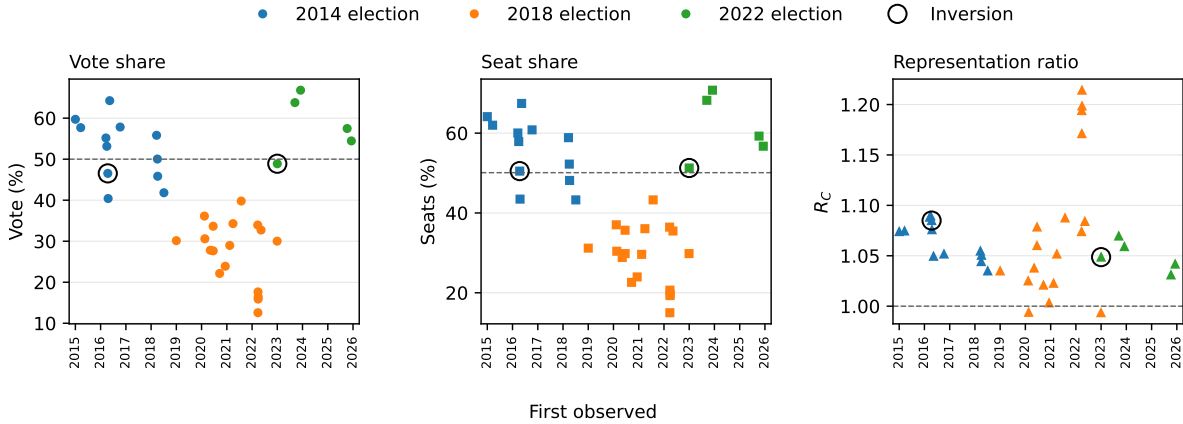


Figure 2: Cabinet party-set vote shares, seat shares, and representation ratios

Inversions occupy 5 of 1,461 covered days after 2014, none of 1,461 after 2018, and 255 of 1,174 after 2022, for 260 of 4,096 days in total. The episodes end differently: the Dilma set loses its seat majority, while the Lula set acquires a vote majority. A fixed electoral distribution can therefore sustain different majority relations as cabinet membership changes.

4.2 Ideologically constrained potential coalitions

There are minimal connected winning inversions in all three elections: four of eight minimal connected winning coalitions invert in 2014, one of eight in 2018, and two of four in 2022. These coalitions are constructed over 28, 30, and 23 seat-winning parties, respectively. The counts

establish that inversions occur in each election; the figures below show where those cases lie in the ideological ordering and how they relate to the wider set of connected majorities. Figure 3 identifies the minimal inversion intervals themselves, while Figure 4 places them within the full exact-connected interval space.

Figure 3 marks PT, the principal organizational and electoral pole of the Brazilian left, as a common reference point. This distinguishes connected majorities extending through PT from those assembled entirely to its right. The four 2014 connected inversions contain twelve to fifteen parties. PSB–PTN begins on the center-left; the other three begin at parties classified as center-right, and all extend into the right. None includes PT, despite PT holding the presidency under Dilma Rousseff and anchoring the left side of the party system. The connected inversion region instead begins to its right: every case combines PMDB and PSD with PSDB. The strongest inversion, measured by the lowest vote share, is the thirteen-party PTB–PR interval: 47.45 percent of all valid votes yields exactly 257 seats, a quota of 243.40 seats, and a differential of 13.60 seats.

The 2018 configuration differs. PT–PSDB is the only exact-connected minimal inversion, so the inversion interval now extends from the principal party of the left through PSDB rather than excluding PT altogether. Its fourteen parties receive 49.95 percent of all valid votes and hold 267 seats. Figure 3 makes an important distinction visible: this is a broad interval in terms of the number of party positions it spans, but its vote share lies only 0.05 percentage points below one half. The proportional quota is 256.25 seats and the differential is 10.75 seats. The other seven minimal connected majorities have vote majorities, including both overrepresented and underrepresented combinations. This election therefore supplies a potential inversion very close to the vote-majority threshold under exact connectedness, even though no cabinet party set linked to the same election inverts in the primary chronology.

In 2022, the non-inverted minimal majorities PT–PP and PSB–PSC begin on the left and center-left and contain fifteen and sixteen parties, respectively. They retain seat majorities despite negative differentials of 9.36 and 15.10 seats. The inversions occur in the ten-party MDB–UNIÃO interval, spanning the center-right through the far right, and the seven-party PP–PL interval, confined to the right and far right.

MDB–UNIÃO receives 49.22 percent of the vote, placing its proportional quota at 252.49 seats, 4.51 below the majority threshold. Its 12.51-seat differential yields 265 seats. PP–PL receives 45.26 percent of the vote and has a quota of 232.19 seats, leaving a 24.81-seat shortfall from 257. Its larger differential of 25.81 seats yields a smaller majority of 258 seats. PL and UNIÃO contribute 13.78 and 11.10 seats above quota to this latter coalition, which also includes underrepresented members. The contrast shows why the size of a seat majority cannot be read directly from the extent of overrepresentation: a coalition with less electoral support requires a larger advantage to reach the same threshold. All baseline $k = 0$ minimal winning intervals appear in Appendix A.2.

Figure 3 shows that the seven minimal inversions differ substantially in ideological location, span, and electoral support. The four 2014 cases form a set of overlapping center-to-right intervals, while the sole 2018 inversion is a broad fourteen-party interval from PT to PSDB whose vote share, 49.95 percent, lies only marginally below one half. The 2022 cases again lie entirely to PT’s right. Most notably, PP–PL reaches 258 seats with only seven adjacent parties and 45.26 percent of the vote, making it both the most compact 2022 inversion and the case with the largest vote deficit among the seven minimal inversions. MDB–UNIÃO occupies a broader center-right-to-far-right span and lies much closer to a vote majority.

Figure 4 shows whether these minimal cases remain inverted as their connected intervals expand. Blue cells have both vote and seat majorities, red cells are minimal connected inversions, and orange cells are larger connected coalitions that remain inverted. Whereas Figure 3 identifies the

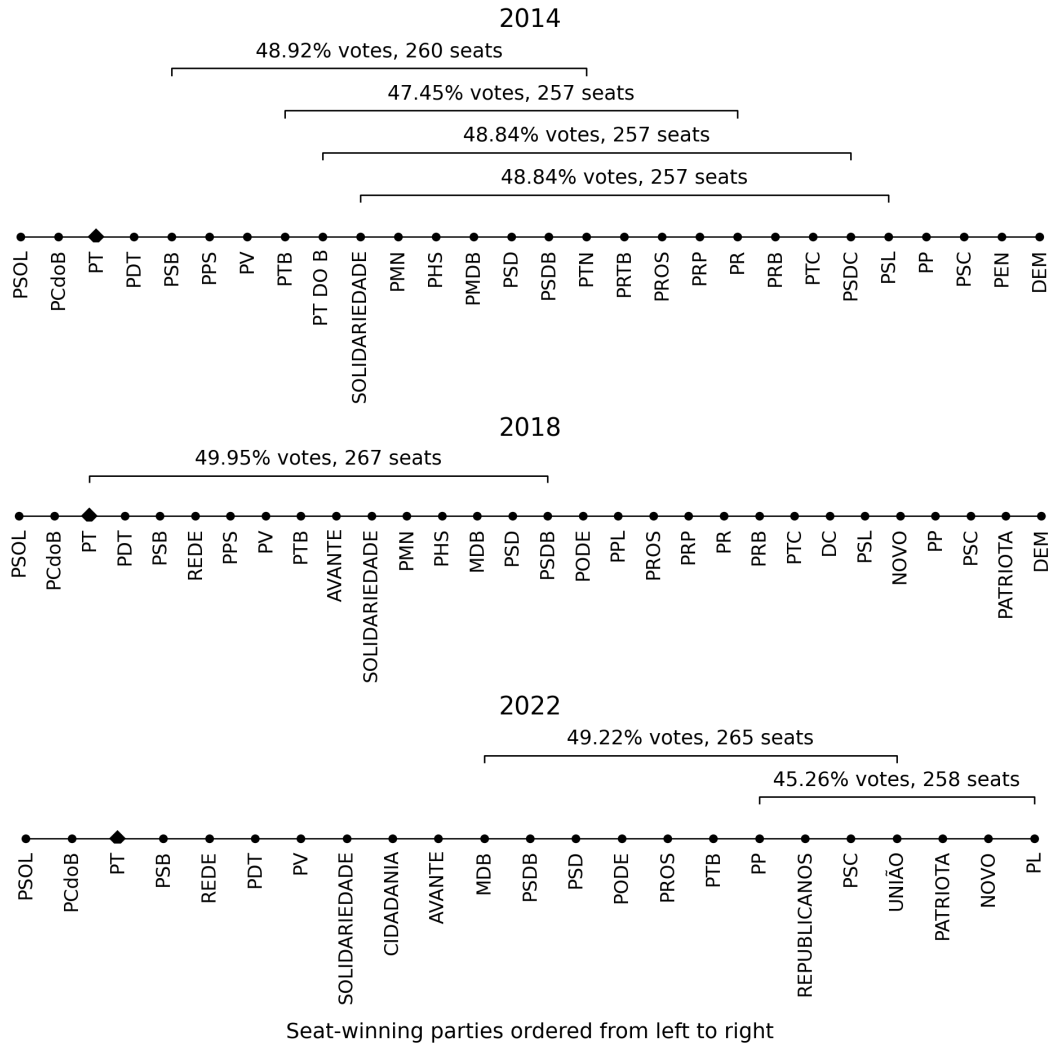


Figure 3: Minimal connected winning inversions by election year. Brackets identify coalition endpoints and report vote shares and seats; the diamond marks PT as a common reference point.

membership and endpoints of the minimal cases, the heatmaps show how those cases sit within the surrounding space of connected coalitions.

The structure differs substantially across elections. In 2014, several minimal inversions lie in overlapping center-to-right spans, and some are accompanied by orange cells. The inversion is therefore not confined to the minimal winning coalitions themselves: some larger connected intervals also retain a seat majority while remaining below 50 percent of the national vote. In 2018, the blue region of vote-and-seat majorities contracts and its boundary shifts toward intervals extending farther to the right of the party ordering. At the same time, PT–PSDB is the only inversion and there are no orange cells. This is a distinct feature of the 2018 configuration. The PT–PSDB interval has 49.95 percent of the vote and 267 seats, but any larger exact-connected interval containing it also has a vote majority. The single inversion is therefore isolated within the surrounding connected coalition space rather than forming part of a larger set of inverted intervals.

The 2022 pattern differs again. The blue region is smaller, reflecting a lower proportion of connected coalitions with both vote and seat majorities, while the inversion cells are concentrated on the right. Unlike 2018, orange cells reappear: some connected expansions of the minimal right-side inversions continue to hold a Chamber majority without reaching 50 percent of the national vote. Connected majorities extending through PT, by contrast, require broader intervals and have both vote and seat majorities. The 2022 result is therefore not merely the existence of two minimal inversions, but an asymmetry in where connected vote-and-seat majorities and connected vote-minority seat majorities occur within the party ordering.

Figure 3 and Figure 4 thus reveal different aspects of the same result: the first identifies the coalitions at the minimal winning boundary, while the second shows whether inversion remains present as the connected interval is expanded. The right-side concentration in 2022 is consistent with the party-level results above, where large right and center-right parties, particularly PL and UNIÃO, carry substantial positive seat differentials that can reinforce one another when adjacent parties are combined.

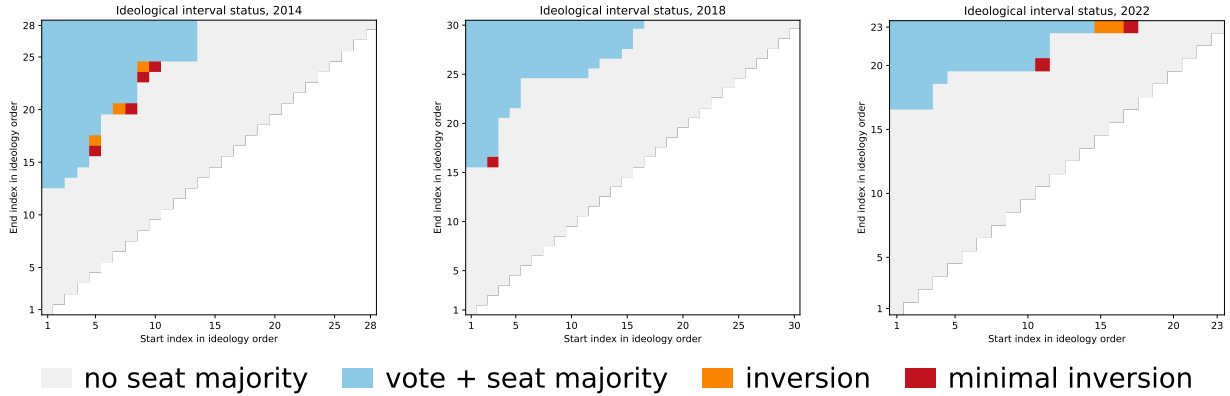


Figure 4: Exact-connected ($k = 0$) ideological interval status by election

Note: Start and end indices refer to the election-year ideological order of seat-winning parties.

Allowing one represented party inside a coalition’s ideological span to be omitted ($k = 1$) does not overturn the main pattern: minimal inversions remain present in all three elections. The strongest 2014 and 2022 cases retain the same member sets and vote–seat outcomes. In 2018, the relaxation produces a stronger inversion than exact connectedness: PCdoB–PODE omitting PSDB, with 47.59 percent of the vote and 257 seats. Appendix B.1 reports the formal definition and summary one-gap robustness results; full coalition-level results are in the machine-readable replication outputs. A separate sensitivity to the ideological party universe preserves the strongest 2014 and 2022 regions but removes the 2018 exact-connected inversion when parties without seats are restored to the ideological order; Appendix B.2 reports the comparison.

4.3 Comparing decomposition patterns across coalition domains

The inversions identified in the two coalition domains share a positive within-district component, but differ in how between-district weighting contributes to their seat majorities. Table 2 reports the decomposition for the minimal connected inversions, complementing the cabinet results in Table 1. Reading the tables together separates two questions: which component accounts for most of a coalition’s seat differential, and whether that component is sufficient to cover the distance from its proportional quota to 257 seats. A component can account for most of the differential without being sufficient to produce the seat majority.

Table 2: Minimal connected winning inversions and seat-differential decomposition

Election	Start	End	Parties	Vote %	Seats	d_C	A_C	B_C
2014	PSB	PTN	12	48.92	260	9.032	10.509	-1.477
2014	PTB	PR	13	47.45	257	13.601	12.507	1.094
2014	PT DO B	PSDC	15	48.84	257	6.442	7.153	-0.711
2014	SOLIDARIEDADE	PSL	15	48.84	257	6.462	7.473	-1.011
2018	PT	PSDB	14	49.95	267	10.751	6.568	4.183
2022	MDB	UNIÃO	10	49.22	265	12.507	2.953	9.554
2022	PP	PL	7	45.26	258	25.813	23.321	2.492

The A_i contributions of individual coalition members help explain the persistent positive within-district component of cabinet coalitions.³ In the 2016 cabinet inversion, PR and PSD contribute 5.04 and 4.65 seats to A_C . In the initial 2023 cabinet, UNIÃO and PT contribute 9.36 and 9.07 seats. Cabinet membership repeatedly brings substantial positive A_i contributions into the same coalition. As the party-size comparison in Section 4.1 shows, large members usually supply enough positive contributions to offset the negative ones, but smaller members also matter. The resulting A_C is positive in all 35 distinct cabinet party sets, including the two inversions.

Cabinet coalitions show no comparable tendency in the between-district component: B_C is positive in thirteen of the 35 distinct cabinet party sets. The two components can also pull in opposite directions for the same party. In the initial 2023 cabinet, for example, PT contributes 9.07 seats to A_C but -2.13 seats to B_C . Among the 2014 inversions, between-district weighting reinforces the within-district component in the Dilma cabinet and the PTB–PR interval, but offsets it in the other three minimal connected inversions. For the initial Lula cabinet, it also offsets the within-district gain.

The 2018 election permits a comparison between an observed cabinet minority and a potential connected inversion. Cabinet party set 18-11 combines $A_C = 21.41$ with $B_C = -3.55$, but its 222

3. The largest member-party A_i and B_i contributions for the selected cases are reported in Table 5 in Appendix A.3.

seats do not constitute a majority. The PT–PSDB connected inversion instead combines positive contributions from both components, $A_C = 6.57$ and $B_C = 4.18$. Its proportional quota lies only 0.75 seats below the majority threshold, so either component individually exceeds that shortfall. The cabinet’s larger within-district gain therefore does not produce the same majority status: its quota lies much farther below 257 seats. The same electoral result thus permits a cabinet minority with a larger within-district gain and a connected seat majority with a much smaller proportional shortfall.

MDB–UNIÃO therefore provides the exceptional case within the minimal connected winning inversions: it is the only case in which between-district weighting accounts for more of the coalition’s seat surplus than within-district allocation. Its 12.51-seat differential consists of $A_C = 2.95$ and $B_C = 9.55$. The relatively small A_C results from substantial cancellation among its members: positive A_i contributions from UNIÃO and MDB are largely offset by negative contributions from PTB and PODEMOS (Table 5 in Appendix A.3). Its positive B_C , by contrast, indicates that the coalition’s support is distributed relatively toward districts with greater electoral weight. Since its proportional quota of 252.49 seats lies only 4.51 seats below a majority, the between-district component alone is larger than that gap. MDB–UNIÃO is therefore the only inversion whose seat differential is primarily associated with favorable between-district weighting rather than within-district allocation.

PP–PL shows why the component that accounts for most of a coalition’s seat differential need not, by itself, span the distance from its proportional quota to 257 seats. Its $A_C = 23.32$ accounts for most of its 25.81-seat differential, while $B_C = 2.49$ is comparatively small. Yet the coalition’s proportional quota is only 232.19 seats, leaving it 24.81 seats short of a majority. The within-district component therefore falls short of that distance on its own; adding the positive between-district component brings the observed total to 258 seats. The Dilma cabinet inversion and the 2014 PTB–PR interval have the same structure: A_C is the larger component, but both positive components enter the differential that carries the coalition across 257.

The sensitivity analysis for ideological coalitions separates two features that coincide in most of the cabinet and minimal-connected winning cases. Positive within-district contributions remain pervasive: $A_C > 0$ in ninety-seven of the one hundred minimal inversions under the one-gap relaxation. But within-district allocation is not always the larger component. $A_C > B_C$ in seventy-seven of those one hundred cases, with substantial variation across elections: the relationship holds in forty of forty-six inversions in 2014 and thirty-two of forty-two in 2018, but only five of twelve in 2022 (Appendix B.1). Thus, the predominance of within-district gains remains strong overall, while the 2022 inversions reveal a different coalitional pattern in which favorable between-district weighting often provides the larger contribution to the seat surplus.

Restoring parties without seats to the ideological order reinforces the need to distinguish these two patterns. MDB–UNIÃO then combines $A_C = -1.29$ with $B_C = 9.88$ and remains inverted (Appendix B.2). The decomposition therefore shows both a persistent tendency for inverted coalitions to accumulate within-district gains and meaningful variation in whether those gains or between-district weighting account for the larger share of the resulting seat differential.

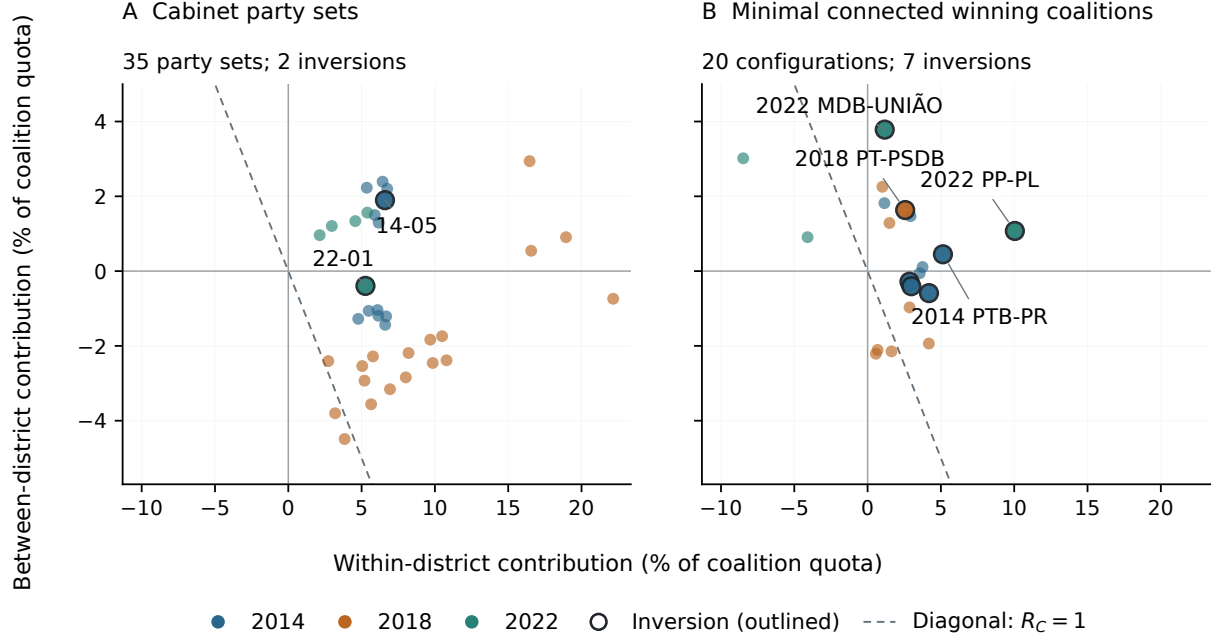


Figure 5: Within-district and between-district contributions across coalition domains

Figure 5 places the inversions within a broader comparison of coalition representation. Panel A includes all 35 distinct election-year cabinet party sets, including those observed under the provisional affiliation assumptions described in Section 3.2. Panel B includes all twenty minimal connected winning coalitions in the primary ideological domain. Both panels express the components relative to each coalition’s proportional quota: the horizontal coordinate is $100A_C/q_C$, and the vertical coordinate is $100B_C/q_C$. Their sum is $100(R_C - 1)$, so coalitions with the same representation ratio lie along parallel lines with slope -1 , even when their component balances differ. The dashed diagonal marks $R_C = 1$, not an inversion boundary: inversion additionally requires less than half of the national vote and at least 257 Chamber seats.

All cabinet sets in Panel A lie to the right of $A_C = 0$, but this shared within-district advantage combines with different patterns of between-district weighting. The 2014 election provides a particularly clear comparison. All six Dilma sets have $B_C > 0$, whereas all six Temer sets have $B_C < 0$. They therefore occupy opposite sides of the horizontal axis despite drawing on the same election-year party votes and seats. This separation reflects cabinet recomposition rather than a change in the electoral outcome: the parties brought together under Temer retain a positive within-district component, but their combined between-district contribution offsets rather than reinforces it.

The Bolsonaro cabinet sets extend farther into the negative between-district region. Fifteen of the eighteen sets linked to the 2018 election have $B_C < 0$, so within-district gains are repeatedly offset by the distribution of their members’ support across districts with different numbers of seats per valid vote. The five Lula sets linked to 2022 mostly occupy the other side of this comparison: four have $B_C > 0$, with the initial cabinet as the exception. These differences in component balance need not reverse overall representation. Across Panel A, 33 of the 35 sets still have $R_C > 1$. Only

the two underrepresented sets linked to 2018 have negative between-district contributions large enough to outweigh their positive within-district gains.

The comparison with Panel B shows that the cabinet patterns do not characterize every politically defined combination available in the same election. Unlike the cabinet sets, minimal connected winning coalitions appear on both sides of $A_C = 0$. Their between-district contributions also differ. In 2018, for example, the PT–PSDB connected inversion combines $A_C = 6.57$ with $B_C = 4.18$, placing it in the upper-right region rather than the lower-right region occupied by most Bolsonaro cabinet sets. The contrast therefore concerns not only whether a coalition holds a majority, but also how the contributions of its members combine. The same electoral distribution permits a connected inversion with two positive components alongside cabinet minorities whose within-district gains are partly offset by negative between-district weighting.

The 2022 cases reveal further variation within and across the domains. All four minimal connected majorities have $B_C > 0$, unlike the initial Lula cabinet. Yet this favorable between-district contribution does not give all four the same representational position. PT–PP and PSB–PSC have negative A_C large enough to leave them underrepresented overall, whereas MDB–UNIÃO and PP–PL receive positive contributions from both components and invert. Even these two inversions differ sharply. Among the baseline cabinet and minimal connected inversions, MDB–UNIÃO is the only case in which the between-district component exceeds the within-district component: $B_C = 9.55$ seats compared with $A_C = 2.95$. PP–PL instead obtains most of its 25.81-seat differential from $A_C = 23.32$, supplemented by $B_C = 2.49$. A common inversion status thus accommodates different balances between the two components.

The initial Lula cabinet and MDB–UNIÃO show how much of this variation can disappear in the aggregate representation ratio. Their ratios are almost identical, approximately 1.049 and 1.050, respectively, but their positions in Figure 5 are not. For the cabinet, A_C/q_C and B_C/q_C are approximately 5.26 and -0.40 percent; for MDB–UNIÃO, they are 1.17 and 3.78 percent. The cabinet’s larger within-district gain is partly offset by between-district weighting, while the connected coalition’s smaller within-district gain is reinforced by it. Together with the Dilma–Temer and 2018 comparisons, this example shows why neither election year nor R_C alone specifies the balance between components. Cabinet membership and ideological connectedness identify different combinations of parties within the same electoral distribution, and those combinations can differ both in their majority status and in the composition of their representational advantage.

5 Discussion

The first result is a contrast between realized rarity and a broader range of potential inversions. Among the 35 distinct cabinet party sets, only two invert. Among the twenty minimal connected winning coalitions, seven do. Cabinet participation therefore shows that coalition inversion was institutionally realized, but only in a small number of the governing party configurations actually observed. The ideological analysis shows, in turn, that these cabinets do not exhaust the politically structured combinations under which the same electoral distributions can reverse vote and seat majorities. The two domains answer different questions: the cabinet analysis identifies inversions that occurred in government, whereas the connected coalition analysis identifies inversion possibilities among ideologically ordered parliamentary majorities.

The relaxation of exact connectedness reinforces this interpretation. Allowing one represented party inside an ideological span to be omitted preserves the strongest inversions in 2014 and 2022 and reveals a stronger case in 2018, where a coalition with 47.59 percent of the vote reaches exactly 257 seats. The existence of potential inversions is therefore not an artifact of requiring every

intervening party to participate. At the same time, these admissible coalitions are not predictions of coalition formation. Rarity among observed cabinets does not establish scarcity among potential coalitions, just as the existence of additional potential inversions does not establish their likelihood of formation.

A second implication is that inversion is not simply a stronger form of overrepresentation. Overrepresentation is common among cabinet coalitions: 33 of the 35 sets have $R_C > 1$, yet only two combine a vote minority with a seat majority. R_C measures the coalition’s advantage relative to its vote share, while d_C expresses that advantage in seats. Whether this advantage changes majority status also depends on q_C , which determines how far the coalition lies from 257 seats under exact proportionality. A positive differential is therefore insufficient for inversion unless it is large enough to bridge that remaining gap, the remaining distance to a seat majority. MDB–UNIÃO and PP–PL illustrate the point. PP–PL is more strongly overrepresented but starts farther below 257, so its larger differential produces the smaller seat majority. Overrepresentation alone therefore determines neither whether a coalition inverts nor the size of the resulting majority.

The Brazilian case adds a second stage to the aggregation problem identified by Kurrild-Klitgaard and Miller. At the coalition level, positive and negative party differentials reinforce or offset one another as parties are combined. In Brazil, however, those party differentials are themselves national aggregates of twenty-seven independently determined district outcomes. The decomposition distinguishes the part of the coalition differential associated with departures from proportionality within districts from the part arising because the sum of district-level proportional quotas differs from the coalition’s national quota. Both components carry into national party seat totals and, ultimately, into coalition majority status.

The analysis therefore follows two successive aggregations. District outcomes are first combined into national party seat totals, and those party-level differentials are then recombined as different political coalitions are formed. The A_C and B_C decomposition adds a separate accounting step by locating the resulting coalition differential within and between districts, while the party contributions show which members add to or offset it. Neither exercise identifies the independent effects of district magnitude, eligibility rules, electoral pooling, malapportionment, or the other mechanisms that contribute to these quantities. The Brazilian extension therefore studies how the established inversion problem operates when national representation is itself produced through multiple district allocations and subsequently recombined at the coalition level.

The decomposition also places the inversion cases within a broader set of coalition configurations. This matters because some of the patterns observed among inverted coalitions are not distinctive to inversion itself. All 35 distinct cabinet party sets have a positive within-district component, whereas the between-district component is positive in only thirteen. The two cabinet inversions therefore belong to a much wider pattern in which cabinet composition repeatedly combines positive within-district contributions, including in party sets that remain minorities in both votes and seats or majorities in both. The party-level contributions help explain this regularity: larger cabinet members often supply substantial positive A_i contributions, while smaller parties can reinforce or offset them. What distinguishes the inverted cabinet sets is consequently not the presence of a positive A_C , but the combination of these components with a coalition quota close enough to the majority threshold for the resulting differential to change majority status.

The ideological coalitions show that the composition of this advantage also varies substantially among inversions themselves. Under the one-gap relaxation, $A_C > 0$ in 97 of the 100 minimal inversions, but $A_C > B_C$ in only 77. In 2022, the within-district component is larger in only five of twelve cases. A positive within-district contribution is therefore pervasive without being universally dominant. Component size must also be distinguished from its role in reaching the

majority threshold. A relatively small positive B_C can provide the remaining seats needed after A_C has brought a coalition close to 257, even when A_C accounts for most of the total differential. This is an arithmetic property of the observed coalition decomposition rather than a counterfactual claim about removing any particular electoral mechanism. The decomposition thus shows that coalitions with the same inversion status, and even coalitions with similar overall representational advantages, can occupy quite different positions with respect to within-district allocation and between-district weighting.

This result deepens Miller’s aggregation argument in an important respect. A fixed electoral distribution does not imply a fixed political beneficiary of coalition inversion. The same party vote and seat totals can sustain different majority relations across governing configurations and can place potential inversions in different regions of the party system. The Dilma inversion ends when the remaining cabinet parties lose their seat majority, whereas the initial Lula inversion ends when the expanded cabinet acquires a vote majority as well. Neither transition reflects a new electoral allocation. The same party vote and seat totals instead support different coalition-level majority relations at different moments of the legislative term.

The ideological results show that this variation is not only temporal. In 2022, the realized Lula cabinet inversion is a broad, sparse, left-anchored coalition (Figure 6), while the minimal connected inversions are located on the right, including the compact PP–PL interval. The same electoral distribution therefore supports inversion in politically quite different parts of the coalition space. The 2018 election provides the converse pattern: no observed cabinet party set inverts, even though an exact-connected inversion exists and the one-gap relaxation reveals a stronger one. Party-level overrepresentation consequently does not imply a stable advantage for one governing bloc or ideological side. Large parties can contribute substantial positive or negative differentials, but which political combinations benefit from them depends on how those contributions are distributed across coalitions. The analysis identifies these representational consequences of coalition structure; it does not imply that parties select partners in order to exploit them.

These results concern a specific relationship between electoral outcomes and politically relevant combinations of parties. Cabinet membership identifies the parties represented in government, not the coalition supporting the executive in every legislative vote. Likewise, the vote share attributed to a cabinet or potential coalition is obtained by summing the election-year votes of its member parties; it is not a direct electoral endorsement of the later coalition as a collective actor. The ideological analysis imposes a different restriction: admissible coalitions are defined relative to a fixed estimated election-year party order, so alternative party placements can change the boundaries of connected or one-gap coalitions. Finally, the $A_C + B_C$ decomposition separates the observed coalition differential into a within-district allocation component and a between-district weighting component. It does not identify the independent causal effects of district magnitude, thresholds, electoral pooling, malapportionment, turnout, or the other institutional and geographical processes that contribute to those components.

The cabinet chronology retains the corresponding election-year vote and seat distribution as its benchmark while cabinet membership changes over the legislative term. It therefore tracks the representational consequences of changing government party sets against a fixed electoral allocation, rather than subsequent changes in parliamentary affiliation or legislative support. These boundaries define the object of the analysis: how national votes and realized seat totals relate to the legislative majorities available to observed and ideologically structured combinations of parties.

6 Conclusion

The Brazilian case shows that coalition inversion is neither merely a theoretical possibility nor a routine feature of coalition politics under proportional representation. It occurs among observed governing party sets, but rarely: only two of the 35 distinct cabinet configurations invert. The same electoral distributions, however, permit a broader range of inversions among ideologically structured parliamentary majorities. Seven of the twenty minimal connected winning coalitions invert, with cases in all three elections, and allowing a single omission inside the ideological span expands that number to 100 of 234 minimal winning coalitions. The central empirical result is therefore a contrast between realized rarity and a wider set of politically structured inversion possibilities.

The analysis also clarifies what makes these cases institutionally consequential. Coalition inversion is not simply another name for overrepresentation: it occurs when the accumulated party-level seat differential is large enough to change majority status. The multidistrict decomposition then distinguishes coalitions whose aggregate advantage is similar in size but composed differently across within-district allocation and between-district weighting. This comparison also reveals a broader pattern beyond the inverted cases themselves. Every observed cabinet party set has a positive within-district component, whereas positive between-district contributions are much less systematic. The inversion cases can therefore be situated within a wider structure of coalition representation, showing both what they share with other party combinations and what makes their accumulated advantage sufficient to alter the vote-majority and seat-majority relation.

Proportional representation is usually described through the relationship between each party's votes and seats. Once those allocations are combined into parliamentary majorities, however, an additional structure becomes visible. Coalition inversion identifies the configurations in which ordinary party-level deviations are sufficient to reverse the relation between vote majorities and seat majorities. This structure makes it possible to ask where such reversals lie within a party system, whether they are realized in governing coalitions, and how their incidence changes as the same legislature is recombined. At the same time, the relation being reversed is a basic one for democratic representation: a set of parties supported by less than half of the national vote can control a majority of legislative seats. Coalition inversions therefore reveal both an analytical property of proportional systems and a majoritarian tension within the parliamentary outcomes those systems can produce.

Statements and Declarations

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Author contribution. The author is the sole author and was responsible for conceptualization, methodology, formal analysis, software, investigation, data curation, visualization, and writing.

Data availability. For peer review, the data and replication materials are available in an anonymized repository: <https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md>. For final publication, the analysis uses public election data from Brazil's Tribunal Superior Eleitoral and cabinet/party classification inputs documented in the replication repository.

Code availability. For peer review, the code necessary to reproduce the analysis is available in an anonymized repository: <https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md>. For final publication, the replication code is available at the public repository after

acceptance.

Ethics approval. This study uses publicly available aggregate electoral, cabinet, and party-classification data. No new human participants were recruited by the author.

Consent to participate. Not applicable. This study uses publicly available aggregate and documentary data.

Consent for publication. Not applicable. This study uses publicly available aggregate and documentary data.

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A Supporting coalition data

A.1 Cabinet membership

Table 3: Distinct election-year cabinet party sets

Set	Election-year member parties	Days
<i>Dilma Rousseff (2014 election)</i>		
14-01	PCdoB, PDT, PMDB, PP, PR, PRB, PROS, PSD, PT, PTB	77
14-02	PCdoB, PDT, PMDB, PP, PR, PRB, PSD, PT, PTB	365
14-03	PCdoB, PDT, PMDB, PP, PR, PROS, PSD, PT, PTB	13
14-04	PCdoB, PDT, PMDB, PP, PR, PSD, PT, PTB	14
14-05	PCdoB, PDT, PMDB, PR, PSD, PT, PTB	5
14-06	PCdoB, PDT, PMDB, PR, PT, PTB	23
<i>Michel Temer (2014 election)</i>		
14-07 [†]	DEM, PMDB, PP, PPS, PR, PRB, PSB, PSD, PSDB, PTB, PV	520
14-08 [†]	DEM, PMDB, PP, PPS, PR, PRB, PSD, PSDB, PTB, PV	158
14-09	DEM, PMDB, PP, PR, PRB, PSD, PSDB, PTB, PV	12
14-10	DEM, PMDB, PP, PRB, PSD, PSDB, PTB, PV	4
14-11	PMDB, PP, PRB, PSD, PSDB, PTB, PV	90
14-12 [†]	PMDB, PP, PRB, PSD, PSDB, PV	180
<i>Jair Bolsonaro (2018 election)</i>		
18-01	DEM, MDB, NOVO, PP, PSL	406
18-02 [†]	DEM, MDB, NOVO, PP, PSDB, PSL	8
18-03 [†]	DEM, NOVO, PP, PSDB, PSL	78
18-04	DEM, PP, PSDB, PSL	41
18-05	DEM, PP, PSD, PSDB, PSL	1
18-06	DEM, PP, PSD, PSL	93
18-07	DEM, PSD, PSL	81
18-08	DEM, PSC, PSD, PSL	65
18-09 [†]	DEM, PRB, PSC, PSD, PSL	47
18-10 [†]	DEM, PR, PRB, PSC, PSD, PSL	119
18-11	DEM, PP, PR, PRB, PSC, PSD, PSL	239
18-12	DEM, PP, PR, PRB, PSC, PSL	3
18-13	PP, PR, PSC	1
18-14	PP, PR, PRB, PSC	2
18-15	PP, PR, PRB	1
18-16	PP, PRB, PSD	41
18-17 [†]	DEM, PP, PRB, PSD, PSL	233
18-18 [†]	DEM, NOVO, PRB, PSD, PSL	2
<i>Lula III (2022 election)</i>		
22-01	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO	255
22-02	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	807
22-03	MDB, PCdoB, PDT, PP, PSB, PSD, PSDB, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	5
22-04	MDB, PCdoB, PDT, PP, PSB, PSD, PSDB, PSOL, PT, REDE, REPUBLICANOS	9
22-05	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS	98

Note: Days sum actual covered dates across appearances, not first-to-last spans. [†]: set includes provisionally coded days. Full histories and coding details remain in the replication files.

A.2 Baseline minimal winning intervals

Table 4: All baseline minimal connected winning intervals

Election	Interval endpoints	Vote %	Seats	d_C
2014	PCdoB-PMDB	50.17	265	7.63
2014	PT-PSD	54.34	291	12.26
2014	PDT-PSDB	51.80	276	10.25
2014	PSB-PTN*	48.92	260	9.03
2014	PTB-PR*	47.45	257	13.60
2014	PT DO B-PSDC*	48.84	257	6.44
2014	SOLIDARIEDADE-PSL*	48.84	257	6.46
2014	PMDB-PP	51.23	272	9.19
2018	PT-PSDB*	49.95	267	10.75
2018	PDT-PR	50.59	268	8.47
2018	PSB-PRB	51.02	269	7.26
2018	SOLIDARIEDADE-PSL	50.12	262	4.86
2018	PMN-NOVO	50.93	257	-4.29
2018	MDB-PP	54.34	285	6.24
2018	PSD-PSC	50.55	258	-1.34
2018	PSDB-DEM	50.82	257	-3.72
2022	PT-PP	57.18	284	-9.36
2022	PSB-PSC	53.82	261	-15.10
2022	MDB-UNIÃO*	49.22	265	12.51
2022	PP-PL*	45.26	258	25.81

Note: Each interval contains all seat-winning parties between its endpoints in the Figure 3 election-year order. *: inversion, classified before rounding; vote shares use all valid federal-deputy votes.

A.3 Selected member-party contributions

Here $d_i = A_i + B_i$, $A_C = \sum_{i \in C} A_i$, and $B_C = \sum_{i \in C} B_i$. These ex post accounting contributions do not identify separate causal effects; in 2014 and 2018, party attribution also reflects joint-list seat allocation.

Table 5: Selected member-party contributions

Case	Within-district A_i		Between-district B_i	
	Positive	Negative	Positive	Negative
Cabinet	PR (5.04)	PT (-1.22)	PMDB (4.22)	PT (-1.21)
2014/14-05	PSD (4.65)	PCdoB (-0.57)	PDT (1.56)	PR (-0.74)
Cabinet	UNIÃO (9.36)	PSB (-5.55)	UNIÃO (1.75)	PSOL (-2.66)
2022/22-01	PT (9.07)	PSOL (-3.40)	PDT (1.02)	PT (-2.13)
Baseline	UNIÃO (9.36)	PTB (-5.94)	PP (3.17)	PSDB (-0.54)
2022/MDB-UNIÃO	MDB (5.12)	PODE (-5.66)	REPUBLICANOS (2.94)	MDB (-0.40)

Note: Entries are the two largest positive and two most negative member contributions per component, in seats. Baseline denotes the seat-winning $k = 0$ interval. PODE is the election-year label for Podemos.

A.4 Inverted cabinets in ideological order

Figure 6 shows the two cabinet inversions as selective memberships in the same ordinal party orders used in Figure 3. Both include PT and span a wide part of their election-year order, omitting twelve represented parties inside the 14-05 span and eleven inside the 22-01 span. In particular, Lula’s PSOL–UNIÃO span is broad, sparse, and left-anchored, whereas the 2022 minimal connected inversions lie entirely to PT’s right. This comparison concerns the two displayed cabinets.

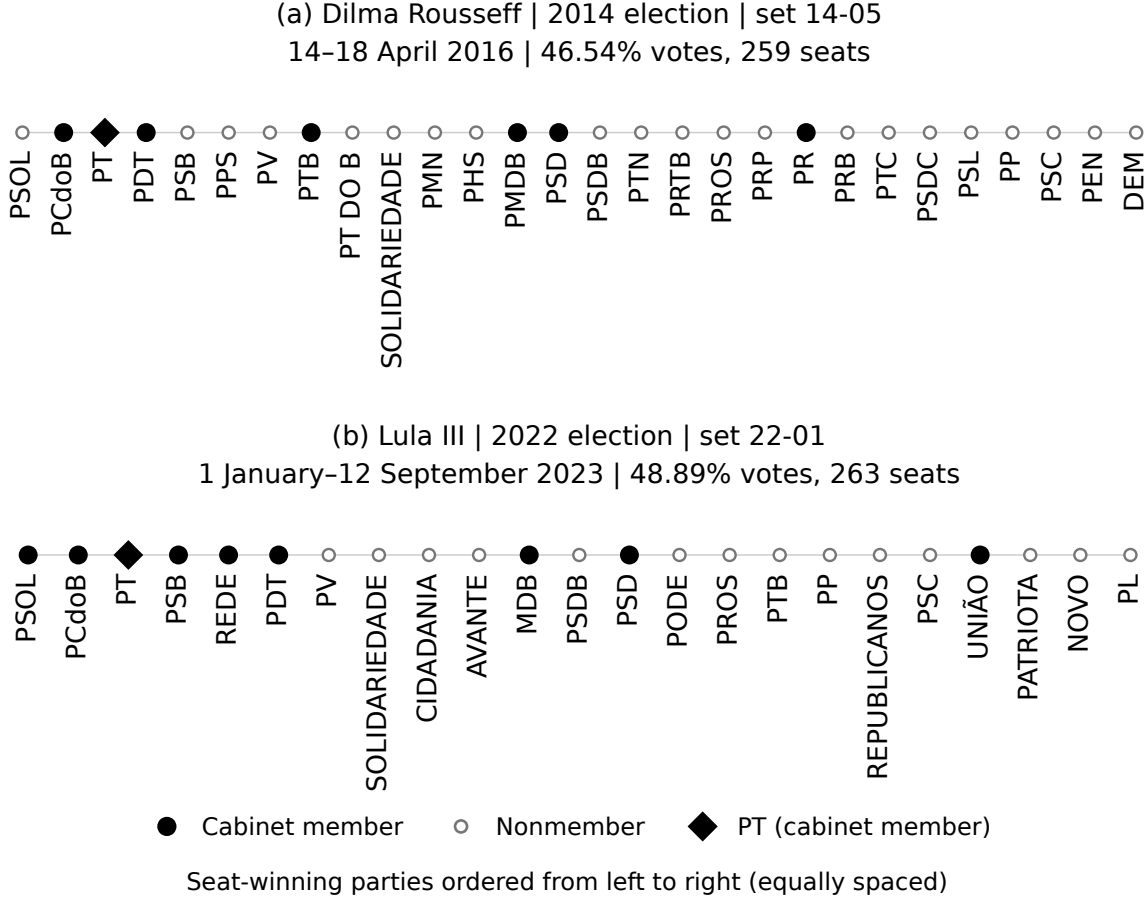


Figure 6: Cabinet inversions in election-year ideological order. Filled markers identify members; open circles identify nonmembers, including omitted parties inside each cabinet’s span. PT is a filled diamond. All 28 seat-winning parties in 2014 and all 23 in 2022 appear at equally spaced ordinal positions. Dates are inclusive observed intervals; votes and seats use the corresponding election.

B Robustness of the ideological coalition domain

B.1 Admissible domains and the one-gap relaxation

Let $p_1, \dots, p_n$ order the parties in P^* , where $P^* = P^+$ contains seat-winning parties in the primary specification and $P^* = P$ contains all parties in the sensitivity analysis. For nonempty $C \subseteq P^*$, let $a = \min\{j : p_j \in C\}$ and $b = \max\{j : p_j \in C\}$. Define

$$\text{span}(C) = \{p_a, \dots, p_b\}, \quad g(C) = |\text{span}(C) \setminus C|, \quad \mathcal{D}_k = \{\emptyset \neq C \subseteq P^* : g(C) \leq k\}.$$

Thus, $k = 0$ requires connectedness, while $k = 1$ permits at most one omitted party inside the span. A winning C is minimal only if no proper subset admissible in the same $\mathcal{D}_k$ retains at least 257 seats. Minimality is recomputed for each k and party universe: nested admissible domains do not imply nested collections of minimal winning coalitions.

All twenty baseline minimal majorities remain admissible and winning under $k = 1$, but only six remain minimal. Table 6 compares the counts. The primary one-gap domain contains 100 inversions among 234 minimal winning coalitions. Full enumeration, memberships, omissions, and decompositions for both universes remain in the machine-readable replication outputs.

Table 6: Minimal winning coalitions by ideological domain and party universe

Election	k	Seat-winning parties		All parties	
		Minimal majorities	Inversions	Minimal majorities	Inversions
2014	0	8	4	8	4
2014	1	87	46	88	43
2018	0	8	1	8	0
2018	1	108	42	118	24
2022	0	4	2	4	2
2022	1	39	12	53	17

Note: Minimality is recomputed for each k and universe. The primary $k = 1$ rows sum to 234 minimal majorities and 100 inversions. The $k = 0$ and $k = 1$ rows are overlapping domains and are not pooled. Every vote share uses all valid federal-deputy votes in V .

The strongest inversion is the one with the lowest national vote share among the relevant minimal inversions. Under the primary one-gap relaxation, the strongest 2014 and 2022 cases retain their baseline member sets and vote-seat outcomes: PTB-PR has 47.45 percent and 257 seats; PP-PL has 45.26 percent and 258 seats. The strongest 2018 case is PCdoB-PODE omitting PSDB, with 47.59 percent of all valid votes and 257 seats.

Among the 100 primary one-gap minimal inversions, $A_C > 0$ in 97 cases and $A_C > B_C$ in 77: 40 of 46 in 2014, 32 of 42 in 2018, and 5 of 12 in 2022. The three negative- A_C cases (two in 2014 and one in 2022) have positive between-district components that more than offset the shortfall.

B.2 Sensitivity to the ideological party universe

Restoring parties without seats removes the 2018 exact-connected PT–PSDB inversion: its interval includes PMB and acquires a vote majority. The strongest 2014 and 2022 endpoint regions persist. Under all-party $k = 1$, their strongest cases omit PPL and DC, respectively, recovering the primary PTB–PR and PP–PL memberships and outcomes. The strongest 2018 all-party one-gap case is PCdoB–PODE omitting PSDB, with 47.82 percent of the vote and the same 257 seats, compared with 47.59 percent in the primary universe.

The national denominator V remains all valid federal-deputy votes in every specification. Changing the ideological universe changes membership, adjacency, coalition vote numerators, and decomposition totals; identical endpoints need not identify identical member sets. In particular, baseline MDB–UNIÃO has $A_C = 2.95$ and $B_C = 9.55$, whereas the all-party interval includes additional seatless members and has $A_C = -1.29$ and $B_C = 9.88$. It remains inverted, with 49.98 percent of all valid votes and 265 seats.